

GENERAL INFORMATION

Name Hilda Sandström
Position Marie Skłodowska-Curie postdoctoral researcher
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CORE COMPETENCES

Scientific leadership · Project management · Molecular modelling & simulation · Structure prediction · Cheminformatics · Machine learning for chemistry · High-performance computing (HPC) · Student supervision & mentoring · Interdisciplinary collaboration · Scientific communication

EXPERIENCE

- Since 10/2025** **Marie Skłodowska-Curie postdoctoral researcher**
Technical University of Munich, Germany
Main project: Machine learning-based compound identification with mass spectrometry
- Develop machine learning models for mass spectrometry signal prediction and dataset similarity analysis.
 - Develop protocols for simulating mass spectrometry data for atmospheric compounds.
 - Coordinate interdisciplinary projects and supervise students.
- 9/2024 – 9/2025** **Visiting postdoctoral researcher**
University of Gothenburg, Sweden
Simulated mass spectrometry signals using machine learning models, molecular dynamics, reaction exploration and quantum chemistry.
- 9/2022 – 9/2025** **Postdoctoral researcher**
Aalto University, Finland
Main project: Machine learning-based compound identification with mass spectrometry
- Developed machine learning models for mass spectrometry signal prediction and dataset similarity analysis.
 - Designed molecular descriptors enabling interpretable machine learning models.
 - Benchmarked models and descriptors for reaction rate prediction.
 - Coordinated interdisciplinary projects and supervised students.
- 9/2017 – 5/2022** **Early-stage researcher (PhD)**
Chalmers University of Technology, Sweden
Main project: Kinetic modeling and molecular structure prediction in polymerization reactions
- Applied steered molecular dynamics, density functional theory, umbrella sampling, and metadynamics for reaction pathway exploration and free-energy profiling.
 - Predicted crystal structures of molecular co-crystals and identified plausible reaction products from kinetics/thermodynamics.
 - Coordinated multi-site collaborations on crystal structure prediction and lipid conformer analysis; advised students.

EDUCATION

- 9/2017 – 5/2022** **PhD in Chemistry (Theoretical chemistry)**
Chalmers University of Technology, Sweden
Award date 02/06/2022. Thesis: *Nitriles in Prebiotic Chemistry and Astrobiology*. Supervisor: Prof. Martin Rahm.
- 8/2012 – 9/2017** **MEng in Chemical engineering with engineering physics**
Chalmers University of Technology, Sweden
Award date 08/11/2017.
- 8/2015 – 9/2017** **MSc in Engineering physics (Nanotechnology master program, integrated)**
Chalmers University of Technology, Sweden
Award date 08/11/2017.
- 8/2012 – 6/2015** **BSc in Chemical engineering with engineering physics (integrated)**
Chalmers University of Technology, Sweden
Award date 12/06/2015.

TEACHING

LECTURES AND EXERCISES

2018–2020	Quantum engineering <i>Chalmers University of Technology</i> Computer labs · 1st year MSc Nanotechnology · 2 h/week
2018–2021	Physical chemistry <i>Chalmers University of Technology</i> Tutorials and experimental labs · 2nd year BSc Biotechnology · 12 h/week
2018–2021	Theoretical chemistry <i>Chalmers University of Technology</i> Computer labs · 3rd year BSc Chemical engineering · 4 h/week
2017–2018	Chemistry and biochemistry <i>Chalmers University of Technology</i> Experimental labs · 1st year BSc Chemical engineering · 8 h/week
2014	Calculus <i>Chalmers University of Technology</i> Exercise sessions · 1st year BSc Chemical engineering · 1 h/week

PEDAGOGICAL TRAINING

2/2026 – 4/2026	Doctoral Supervision Training <i>Joint leadership-skills training — Helmholtz, Max Planck Society, LMU Munich & TUM, Munich</i> 26 Feb – 30 Apr 2026. 32 hours equivalent.
WS 2025/26	Supervising Students' Theses <i>TUM ProLehre, online workshop</i> 9 hours, Category E — Advising & Counseling
2019	Teaching, learning and evaluation <i>Chalmers University of Technology</i> 3 ECTS

SUPERVISION OF STUDENTS

2024–2025	Supervisor of MSc student — Aalto University
Since 2024	Advisor of 2 PhD students — Aalto University
Since 2022	Co-supervisor of PhD student — University of Helsinki
2018–2024	Supervised or co-supervised 22 BSc and visiting undergraduate students — Chalmers University of Technology, Aalto University

PROFESSIONAL DEVELOPMENT

TRAINING

6–7/2026	CSC Summer School in High-Performance Computing 2026 <i>CSC – IT Center for Science, Nuukio, Finland</i> 23 June – 2 July 2026. HPC programming with C++. 5 ECTS or 70 hours.
2/2026 – 4/2026	Doctoral Supervision Training <i>Joint leadership-skills training — Helmholtz, Max Planck Society, LMU Munich & TUM, Munich</i> 26 Feb – 30 Apr 2026. 32 hours equivalent.
3/2025 – 4/2025	Cooking Open Science! <i>Aalto University</i> Blended course on open science principles for doctoral and postdoctoral researchers

LANGUAGE COURSES

10–12/2025	German for Researchers, A1.2 <i>TUM Language Center</i> Grade: 1.3 (very good) · 2 SWS
1–3/2026	German for Researchers, A2.2 <i>TUM Language Center</i> Grade: 2.0 (good) · 2 SWS

SKILLS AND COMPETENCES

PROGRAMMING

Python, MATLAB, Bash — Well experienced

MACHINE LEARNING AND CHEMINFORMATICS

Scikit-learn, TensorFlow, RDKit, OpenBabel, ASE — Experienced

MOLECULAR DYNAMICS AND SIMULATION

CP2K, GROMACS, PLUMED — Expert · xTB, QcXMS, VMD — Experienced

HIGH-PERFORMANCE COMPUTING (HPC)

Parallel computing, cluster resource management — Experienced

VERSION CONTROL

Git — Experienced

LANGUAGES

Swedish (Excellent) · English (Excellent) · Italian (Intermediate) · German (A2) · French (Basic)

AWARDS AND HONOURS

2025	Marie Skłodowska-Curie postdoctoral fellowship 202,000 EUR
2024–2025	LUMI extreme scale access resource allocation
2018–2021	Travel grants Nils Philblad Foundation (2021) · Karl and Annie Leon's Foundation (2018–2019)

ACADEMIC SERVICE

2026	Poster judge — JpGU-AGU Joint Meeting 2026, Chiba, Japan
2026	Thesis reviewer and opponent — Universitat Autònoma de Barcelona (UAB), Barcelona, Spain
2025	Reviewer for <i>ACS Earth Space Chem</i> , <i>ACS Omega</i> and <i>Atmospheric Chemistry and Physics</i>
2025	Organizing committee — Nordic Workshop on AI for Climate Change, Sweden
Since 2025	Core member, organizer and Finland representative — Climate AI Nordics Network; contributes job ads to the monthly newsletter
2024	Panelist on AI in chemistry, physics, and education — FysKemDagarna
2023	Organizer of workshop hands-on session — Shaking Up Tech, Workshop for underrepresented groups in STEM, Aalto University, Finland
2023	Session chair and organizer — ESTML, Levi, Finland
2022	Session chair — AbSciCon, USA

PEER-REVIEWED PUBLICATIONS

15 peer-reviewed articles, 6 first author. Total citations: 157, h-index: 6, i10-index: 5 — Google Scholar, updated 7 July 2026

15. Cappelletti, M., **Sandström, H.**, & Rahm, M. *ACS Central Science*, 12, 111–121 (2026). DOI: 10.1021/acscentsci.5c01497.

14. Lind, L., **Sandström, H.**, & Rinke, P. *The Journal of Chemical Physics*, 164 (2026). DOI: 10.1063/5.0308548.

13. Madan, I., Aliabadi, S. A., Huhtasaari, J., Matic, E., Hogedal, E., Kamińska, K., Nilsson, F., Stark, A., Izquierdo-Ruiz, F., **Sandström, H.**, Rahm, M. *QRB Discovery*, 6, e23 (2025). DOI: 10.1017/qrđ.2025.10012. [Supervised students and co-created workflow for testing stability of polymers.]

12. Brean, J., Bortolussi, F., Rowell, A., Beddows, D. C. S., Weinhold, K., Mettke, P., Merkel, M., Kumar, A., Barua, S., Iyer, S., Karppinen, A., **Sandström, H.**, Rinke, P., et al. *ACS ES&T Air*, 2, 1704–1713 (2025). DOI: 10.1021/acsestair.5c00119. [Supervised PhD student F. Bortolussi in developing and evaluating the machine learning model and workflow.]

11. Izquierdo-Ruiz, F., Cable, M. L., Hodyss, R., Vu, T. H., **Sandström, H.**, Lobato, A., & Rahm, M. *Proc. Natl. Acad. Sci. U.S.A.*, 122, e2507522122 (2025). DOI: 10.1073/pnas.2507522122. [Developed and tested crystal structure prediction program workflow for molecular cocrystals.]

10. Valiev, R. R., Nasibullin, R. T., **Sandström, H.**, Rinke, P., Puolamäki, K., & Kurten, T. *Physical Chemistry Chemical Physics*, 27, 14804–14814 (2025). DOI: 10.1039/D5CP01101A. [Co-advisor for ML workflow; developed MBTR model.]

9. Bortolussi, F., **Sandström, H.**, Partovi, F., Mikkilä, J., Rinke, P., & Rissanen, M. *Atmospheric Chemistry and Physics*, 25, 685–704 (2025). DOI: 10.5194/acp-25-685-2025. [Co-designed study, advised, and contributed to programming and model testing.]

8. Malaska, M. J., **Sandström, H.**, Hofmann, A. E., Hodyss, R., Rensmo, L., van der Meulen, M., Rahm, M., Cable, M. L., & Lunine, J. I. *Astrobiology*, 25, 367–389 (2025). DOI: 10.1089/ast.2024.0125. [Performed geometry optimizations, conformer search and student supervision.]

7. **Sandström, H.**, & Rinke, P. *Geoscientific Model Development*, 18, 2701–2724 (2025). DOI: 10.5194/gmd-18-2701-2025.

6. **Sandström, H.**, Rissanen, M., Rousu, J., & Rinke, P. *Advanced Science*, 11, 2306235 (2024). DOI: 10.1002/advs.202306235.

5. **Sandström, H.**, Izquierdo-Ruiz, F., Cappelletti, M., Dogan, R., Sharma, S., Bailey, C., & Rahm, M. *ACS Earth and Space Chemistry*, 8, 1272–1280 (2024). DOI: 10.1021/acsearthspacechem.4c00088.

4. **Sandström, H.**, & Rahm, M. *The Journal of Physical Chemistry A*, 127, 4503–4510 (2023). DOI: 10.1021/acs.jpca.3c01504.

3. **Sandström, H.**, & Rahm, M. *ACS Earth and Space Chemistry*, 5, 2152–2159 (2021). DOI: 10.1021/acsearthspacechem.1c00195.

2. **Sandström, H.**, & Rahm, M. *Science Advances*, 6, eaax0272 (2020). DOI: 10.1126/sciadv.aax0272.

1. Lindblom, A., Sriram, K. K., Müller, V., Öz, R., **Sandström, H.**, Åhrén, C., Westerlund, F., & Karami, N. *Diagnostic Microbiology and Infectious Disease*, 93, 380–385 (2019). DOI: 10.1016/j.diagmicrobio.2018.10.014. [Performed fluorescence microscopy assays where I stained, trapped, and photographed plasmids in nanochannels.]

TALKS

INVITED SEMINARS AND KEYNOTES

2026	Data-driven mass spectrometry for atmospheric compound identification <i>University of Gothenburg, Sweden</i>
2026	Data-driven compound identification with atmospheric mass spectrometry <i>Network on Mathematical Data Science for Materials Science, Workshop on the Interface of Mathematics and Machine Learning for Applications in Materials Science — University of Glasgow, UK</i>
2025	CLOUDMAP – Advanced identification of atmospheric compounds <i>Atmospheric day, Sweden — Keynote</i>
2025	Machine learning for atmospheric mass spectrometry <i>Nordic Workshop on AI for Climate Change, Sweden</i>
2024	AI in Chemistry: Solving experimental challenges with artificial intelligence <i>FysKemDagarna (Physics and Chemistry Days), Sweden</i>

CONTRIBUTED TALKS

2026	Towards Atmospheric Compound Identification Using Simulated Electron Ionization Mass Spectra <i>JpGU-AGU Joint Meeting 2026, Chiba, Japan</i>
2026	Towards atmospheric compound identification using simulated electron ionization mass spectra <i>Chemical Compounds Space Conference (CCSC 2026), Munich, Germany</i>
2023	Characterizing Atmospheric Molecules for Machine Learning <i>International Aerosol Modeling Algorithms Conference, USA</i>
2023	Characterizing Atmospheric Molecules for Machine Learning <i>European Aerosol Conference, Spain</i>
2023	Characterizing atmospheric molecules for machine learning <i>Physics Days, Finland</i>
2022	Untangling hydrogen cyanide polymerization using quantum chemistry <i>AbSciCon, USA</i>